

# Bhaumik Bhalara

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## SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in Verilog, Python, and FPGA hardware design. Proficient in collaborating with interdisciplinary teams to drive innovative projects in cloud hardware development. Seeking to leverage technical expertise to contribute to groundbreaking advancements in AWS's Generative AI and ML server offerings.

## EXPERIENCE

### Intern

Essar Power Hazira LTD, Surat, India

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems in boiler-actuated power generation, enhancing system reliability by 20%.
- Conducted in-depth research on link mechanisms for transporting waste loads from boilers, resulting in a proposed design that improved efficiency by 15%.
- Monitored and recorded gauge and probe data, ensuring 100% safety compliance during operational seminars and workshops.

### Intern

3D Cubic, Surat, India

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating advanced 3D printing technology, leading to the successful fabrication of a 2 cm x 2 cm x 2 cm polyamide cube.
- Demonstrated hands-on expertise in 3D modeling and printing processes, contributing to a 30% reduction in prototype development time for new designs.

## EDUCATION

### Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

### Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

## SKILLS

Verilog, Python, FPGA Design, CAD Tools, MATLAB, Signal Integrity, Power Management, Thermal Management, Project Management, Team Collaboration, 3D Printing Technology, Hardware Development, System Design

## PROJECTS

### 64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed SRAM with comprehensive block and state diagrams for PCI to SRAM transactions, achieving a 25% increase in transaction speed.
- Prepared detailed timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring compliance with industry standards and specifications.
- Collaborated with a team of 4 engineers to validate design performance through simulation and testing, ensuring a successful deployment in a real-world application.

### Vending Machine System

Aug 2023 - Dec 2023

- Integrated multiple finite state machines including idle, selection, payment, dispensing, and reset states, resulting in a 40% improvement in operational efficiency.
- Tested and verified the design using Verilog testbenches and hardware validation on an Intel DE-10 Lite FPGA board, achieving a 99% accuracy rate in transactions.

- Led a team of 3 in developing the project, enhancing collaboration and project management skills.

### **6-Bit Segmented DAC**

*Jan 2024 - May 2024*

- Implemented DAC design divided into LSB and MSB segments, achieving expected results for both theoretical and simulated values of 12V with less than 5% variance.
- Optimized design through iterative testing, reducing glitch noise by 30% and enhancing overall signal integrity.
- Collaborated with peers to analyze performance metrics, contributing to a comprehensive final report.

### **STA for Microprocessor Netlist**

*Aug 2024 - Dec 2024*

- Generated comprehensive setup and hold timing reports, identifying timing violations and optimizing slack to improve overall design performance by 15%.
- Utilized advanced timing analysis tools to ensure compliance with stringent design specifications, resulting in successful project completion ahead of schedule.
- Collaborated with a multidisciplinary team to refine design processes, leading to improved efficiency in future projects.

### **4-Bit Ripple Carry Adder**

*Aug 2024 - Dec 2024*

- Implemented Verilog module and Perl script to thoroughly verify the functionality of the adder, achieving 100% accuracy in all test cases.
- Generated detailed timing reports using design constraints, identifying critical paths and optimizing performance metrics for static timing analysis.
- Worked with a team of 2 to streamline the design process, improving collaboration and project delivery timelines.